

Budget-Matched Controls Limit Model-Strength and Replay Claims in LLM Code Evolution

Purpose

This factorial analysis tests whether evolutionary code search adds performance above base model coding strength across simple games, symmetric TSP, and real-world CVRP.

Experimental design

Rows analyzed: 750. Task families present: cvrp_phase9_real_world, simple_games, tsp.

The planned full design is 3 model tiers x 5 techniques with 10 simple-game seeds and 20 TSP/CVRP seeds per cell. Partial or smoke outputs are marked by their row counts.

Model tiers and benchmark strength scores

See model_strength_table.csv. The analysis supports both categorical model_tier and continuous benchmark_strength_score.

Evolutionary techniques

Techniques: single_shot, budget_matched_no_replay, random_replay, failure_replay, and failure_replay_compression.

Task-specific results

task	model	technique	n	mean performance_z	mean raw	mean novelty
cvrp_phase9_real_world	medium_model	budget_matched_no_replay	20	0.3708	-8.585	0.6181
cvrp_phase9_real_world	medium_model	failure_replay	20	-0.1425	-11.14	0.2734
cvrp_phase9_real_world	medium_model	failure_replay_compression	20	-0.2991	-11.92	0.3233
cvrp_phase9_real_world	medium_model	random_replay	20	-0.4003	-12.42	0.3095
cvrp_phase9_real_world	medium_model	single_shot	20	-0.1432	-11.14	0
cvrp_phase9_real_world	strong_model	budget_matched_no_replay	20	0.8589	-6.156	0.569
cvrp_phase9_real_world	strong_model	failure_replay	20	1.484	-3.047	0.4093

task	model	technique	n	mean performance_z	mean raw	mean novelty
cvrp_phase9_real_world	strong_model	failure_replay_compression	20	-0.1675	-11.26	0.366
cvrp_phase9_real_world	strong_model	random_replay	20	1.097	-4.974	0.3702
cvrp_phase9_real_world	strong_model	single_shot	20	-0.2717	-11.78	0
cvrp_phase9_real_world	weak_model	budget_matched_no_replay	20	-0.5287	-13.06	0.6193
cvrp_phase9_real_world	weak_model	failure_replay	20	-0.5287	-13.06	0.3648
cvrp_phase9_real_world	weak_model	failure_replay_compression	20	-0.5287	-13.06	0.369
cvrp_phase9_real_world	weak_model	random_replay	20	-0.2709	-11.78	0.3818
cvrp_phase9_real_world	weak_model	single_shot	20	-0.5287	-13.06	0
simple_games	medium_model	budget_matched_no_replay	10	0.5659	0.5273	0.6598
simple_games	medium_model	failure_replay	10	-0.9354	0.1909	0.07102
simple_games	medium_model	failure_replay_compression	10	-0.7812	0.2255	0.0605
simple_games	medium_model	random_replay	10	-0.2294	0.3491	0.04661
simple_games	medium_model	single_shot	10	-0.9029	0.1982	0
simple_games	strong_model	budget_matched_no_replay	10	0.8823	0.5982	0.5775
simple_games	strong_model	failure_replay	10	0.4604	0.5036	0.562
simple_games	strong_model	failure_replay_compression	10	0.6308	0.5418	0.506
simple_games	strong_model	random_replay	10	0.8418	0.5891	0.5432
simple_games	strong_model	single_shot	10	-0.2943	0.3345	0
simple_games	weak_model	budget_matched_no_replay	10	0.6064	0.5364	0.6699
simple_games	weak_model	failure_replay	10	-0.02651	0.3945	0.2251
simple_games	weak_model	failure_replay_compression	10	-0.09954	0.3782	0.1744
simple_games	weak_model	random_replay	10	0.03841	0.4091	0.2129

task	model	technique	n	mean performance_z	mean raw	mean novelty
simple_games	weak_model	single_shot	10	-0.7568	0.2309	0
tsp	medium_model	budget_matched_no_replay	20	0.07828	-0.2134	0.728
tsp	medium_model	failure_replay	20	0.07828	-0.2134	0.06069
tsp	medium_model	failure_replay_compression	20	-0.003437	-0.2136	0.2038
tsp	medium_model	random_replay	20	-0.1953	-0.214	0.05938
tsp	medium_model	single_shot	20	0.07828	-0.2134	0
tsp	strong_model	budget_matched_no_replay	20	-0.7406	-0.2151	0.6953
tsp	strong_model	failure_replay	20	0.07828	-0.2134	0.4104
tsp	strong_model	failure_replay_compression	20	0.07828	-0.2134	0.3182
tsp	strong_model	random_replay	20	0.07828	-0.2134	0.4169
tsp	strong_model	single_shot	20	0.07828	-0.2134	0
tsp	weak_model	budget_matched_no_replay	20	0.07828	-0.2134	0.7468
tsp	weak_model	failure_replay	20	0.07828	-0.2134	0.3393
tsp	weak_model	failure_replay_compression	20	0.07828	-0.2134	0.2511
tsp	weak_model	random_replay	20	0.07828	-0.2134	0.3484
tsp	weak_model	single_shot	20	0.07828	-0.2134	0

Model x evolution matrices

CSV matrices and matching matrix heatmaps are in `model_x_evolution_matrices/`; duplicated report heatmaps are in `figures/`.

Variance decomposition

Formulas:

- Task continuous: $\text{performance_z} \sim \text{benchmark_strength_score} + C(\text{evolution_technique}) + \text{benchmark_strength_score}:C(\text{evolution_technique})$.
- Task categorical: $\text{performance_z} \sim C(\text{model_tier}) * C(\text{evolution_technique})$.
- Pooled continuous: $\text{performance_z} \sim C(\text{task_family}) * \text{benchmark_strength_score} * C(\text{evolution_technique})$.
- Pooled categorical: $\text{performance_z} \sim C(\text{task_family}) * C(\text{model_tier}) * C(\text{evolution_technique})$.

Pooled continuous model: model-strength $R^2=0.0008843$, evolution-technique $R^2=0.01002$, interaction $R^2=0.06909$, residual variance=0.8298.

Effect sizes relative to single-shot

10 comparisons have bootstrap CIs above zero relative to single-shot.

Effect sizes relative to budget-matched no-replay

0 comparisons have bootstrap CIs above zero relative to budget-matched no-replay.

Cross-family interpretation

The interpretation should focus on the budget control. Replay/failure/compression techniques beat budget-matched no-replay in 0 tested task/model comparisons with positive bootstrap support; therefore, gains over single-shot should be interpreted primarily as effects of extra search budget unless a task-specific budget-control comparison supports a stronger claim. Pooled continuous model: model-strength $R^2=0.0008843$, evolution-technique $R^2=0.01002$, interaction $R^2=0.06909$, residual variance=0.8298.

Main conclusion

The budget-feasible model ladder does not show clean model-strength dominance, and replay/failure/compression do not beat the budget-matched no-replay control. Pooled model-strength $R^2=0.0008843$, evolution-technique $R^2=0.01002$, and interaction $R^2=0.06909$; the strongest conclusion is that apparent gains over single-shot mostly reflect search budget and task/model-specific interactions rather than robust replay-specific value.

Limitations

- Do not overclaim algorithmic discovery; this phase compares model strength and search technique, not novelty of discovered algorithms.
- Replay/failure/compression should be claimed useful only when they beat `budget_matched_no_replay`, not merely when they beat `single_shot`.
- Smoke runs validate schema only; scientific conclusions require the full paid campaign.
- P-values use reduced-model OLS/ANOVA approximations implemented locally to avoid adding heavy statistics dependencies.

Required Questions

1. How much variance is explained by model strength? 0.0008843.
2. How much variance is explained by evolutionary technique? 0.01002.
3. Is there a model strength x technique interaction? 0.06909.
4. Do replay/failure/compression techniques beat a budget-matched no-replay control? 0 comparisons beat budget-matched no-replay with positive bootstrap support.
5. Are evolutionary gains larger for weak models than strong models? insufficient or no positive budget-control gains.
6. Is the strongest conclusion model strength dominates, evolution adds independent value, or evolution only helps under certain task/model regimes? The budget-feasible model ladder does not show clean model-strength dominance, and replay/failure/compression do not beat the budget-matched no-replay control. Pooled model-strength $R^2=0.0008843$, evolution-technique $R^2=0.01002$, and interaction $R^2=0.06909$; the strongest conclusion is that apparent gains over single-shot mostly reflect search budget and task/model-specific interactions rather than robust replay-specific value.

Artifact Paths

- all_runs_long.csv: runs\cross_family_model_x_evolution_factorial\20260524_172130\all_runs_long.csv
- variance_partition_summary.csv: runs\cross_family_model_x_evolution_factorial\20260524_172130\variance_decomposition\variance_partition_summary.csv